

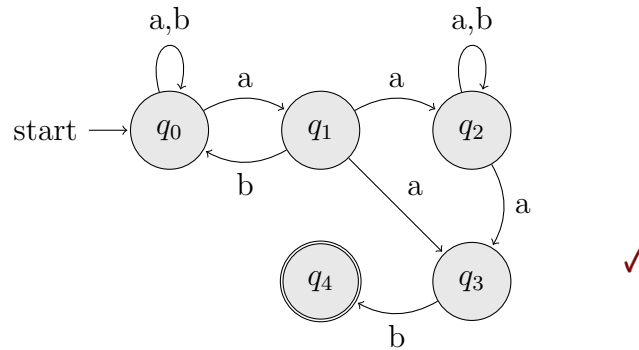
Automaten und Sprachen, Übungsblatt 2

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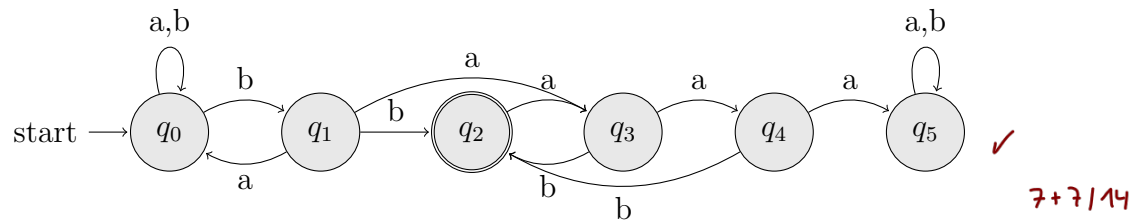
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4. $\mathcal{A}_1$:

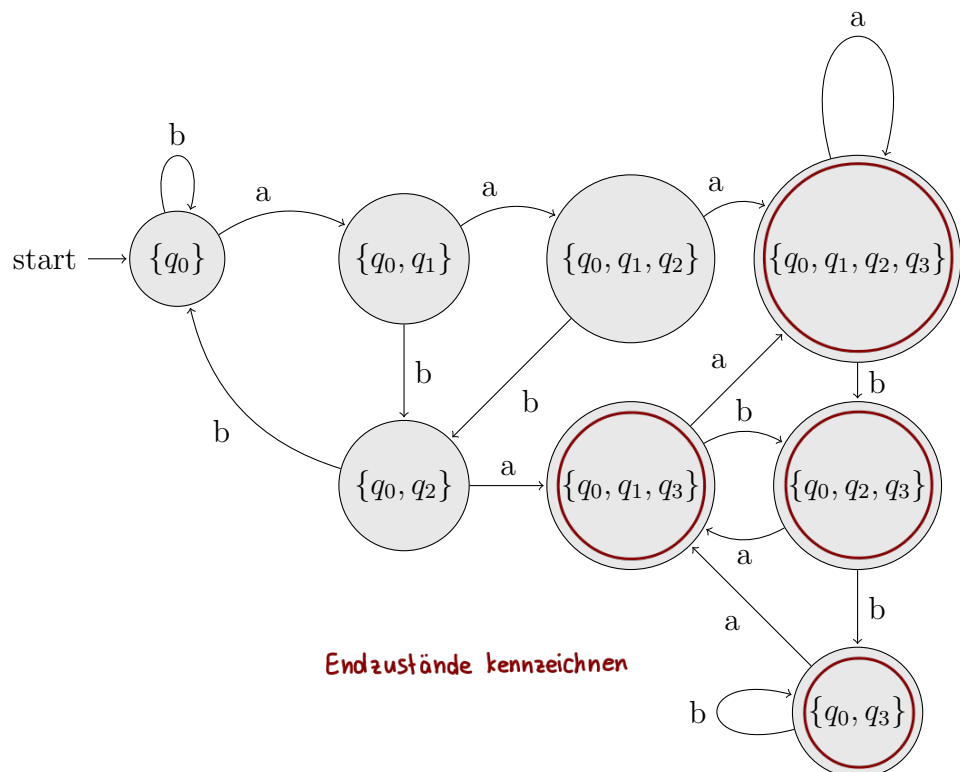


$\mathcal{A}_2$:



5. (a) $L(\mathcal{A}) = \{w \in \{a, b\}^* \mid \exists i \in \{1, \dots, |w| - 2\} : w_i = w_{i+2} = a\}$ ✓

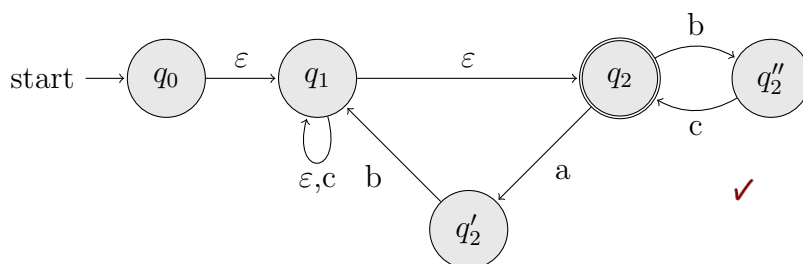
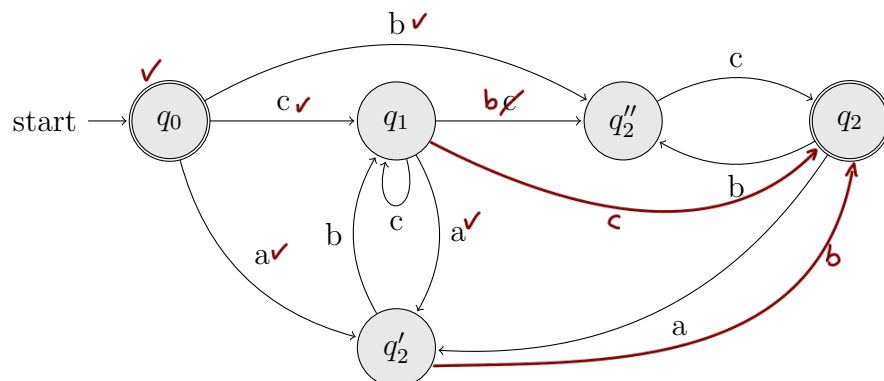
(b) $\mathcal{A}'$:



Endzustände kennzeichnen

	a	b
$\{q_0\}$	$\{q_0, q_1\}$	$\{q_0\}$
$\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_2\}$
$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_2\}$
$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_2, q_3\}$
$\{q_0, q_2\}$	$\{q_0, q_1, q_3\}$	$\{q_0\}$
$\{q_0, q_1, q_3\}$	$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_2, q_3\}$
$\{q_0, q_2, q_3\}$	$\{q_0, q_1, q_3\}$	$\{q_0, q_3\}$
$\{q_0, q_3\}$	$\{q_0, q_1, q_3\}$	$\{q_0, q_3\}$

2 + 9 / 12

6. (a) $\mathcal{A}'$:(b) $\mathcal{A}''$:

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7. Für $n_0 \in \mathbb{N}_+$ beliebig, wähle $w = a^{n_0}b^{2n_0} \in L$. Dann $|w| \geq n_0$.

Sei Zerlegung $w = xyz$ so, dass $|xy| \leq n_0$ und $y \neq \varepsilon$. Dann gibt es $k_1, k_2 \in \mathbb{N}$, $k_2 \geq 1$ so dass $x = a^{k_1}$, $y = b^{k_2}$, $z = a^{n_0-k_1-k_2}b^{2n_0}$. Aber $xy^2z \notin L$, da

$xy^2z = a^{k_1}a^{2k_2}a^{n_0-k_1-k_2}b^{2n_0} = a^{n_0+k_2}b^{2n_0}$ und $2(n_0 + k_2) = 2n_0 + 2k_2 \neq 2n_0$.

Also L nicht erkennbar.

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